

State Machine Design

1. Define the minimum hardware configuration required for each mode.

Safe Mode: Ensures the survival of the ADCS linked to limitation of power consumption (no actuators).

Minimum hardware required:

2 x 3-axis magnetic field sensor (hot redundant)	All sensors are used
2 x 3-axis gyroscope (cold redundant)	in order to calculate
2x 4pi sun sensor assembly (hot redundant)	the most precise
1x GPS receiver	attitude determination

Damping Mode: Any rotation of the satellite has to be stopped.

Minimum hardware required:

2 x 3-axis gyroscope (cold redundant)	To calculate the RPM of the reaction wheels
4 x reaction wheels in tetrahedron configuration	To stop the angular motion of the satellite.
3 x magnetic coils in orthogonal configuration	To desaturate the reaction wheels.

Sun pointing Mode: Points the most part of solar array's surface towards the Sun.

Minimum hardware required:

2x 4pi sun sensor assembly (hot redundant)	To calculate sun's direction.
2 x 3-axis gyroscope (cold redundant)	To calculate the RPM of the reaction wheels
4 x reaction wheels in tetrahedron configuration	To point to the sun.

Earth pointing Mode: Points instruments towards the Earth.

Minimum hardware required:

1x GPS receiver	To complete the Magnetic Field model
2 x 3-axis magnetic field sensor (hot redundant)	To calculate Earth's direction.
2 x 3-axis gyroscope (cold redundant)	To calculate the RPM of the reaction wheels
4 x reaction wheels in tetrahedron configuration	To point to the Earth.

Slew Mode: Points from one direction to another direction

Minimum hardware required:

2 x 3-axis gyroscope (cold redundant)	To calculate the RPM of the reaction wheels
4 x reaction wheels in tetrahedron configuration	To point to another direction

2) Design a state machine containing all above-mentioned modes. List entry and exit conditions and triggers (ground TC or onboard TC)

RW: Reaction Wheels

MC: Magnetic Coils

MFS: Magnetic Field Sensor

